Roberto Emmanuel Valenzuela Armenta

Electronics Engineering Student
https://www.linkedin.com/in/roberto-valenzuela/

+52 1 644 118 5270
roemvaar.github.io
valenzuelarober@gmail.com

Education

- **Instituto Tecnológico de Sonora** (Mexico, 08/2014 - present)
  Electronics Engineering, Bachelor's degree
  Thesis: "Design of a co-processor for the convolution operation with Verilog HDL"
- **Åbo Akademi University** (Finland, 08/2017 - 12/2017)
  Computer Engineering, Exchange student
- **Arizona State University** (USA, 11/2014 - 12/2014)
  American English and Culture Program

Work Experience

- **Instituto Tecnológico de Sonora** (Mexico, 10/2018 - present)
  Undergraduate Research Assistant
  Working as a data scientist on a biostatistics paper researching the spread of Rickettsia disease in Mexico and proposing solutions. Used Python for data wrangling from an unorderdered data source and performed a general data analysis. Applied machine learning algorithms to extract insights from the data. Used Latin Hypercube Sampling to approximate parameters.
- **fileee GmbH** (Germany, 05/2018 - 08/2018)
  Web Developer Intern
  Worked on the main web app project. Responsible to migrate the fileee web app from Aurelia to React using extensively modern JavaScript libraries and tooling (React and ESNext), HTML and CSS, based in the Agile methodology.

Volunteer Experience

- Vice-president of the "Electronics engineering society" (08/2016 - present)
- Instructor for the course "Introduction to data science" (12/2018)
- English, programming and math tutor (08/2014 – 12/2017)
- Data Analyst Intern at Business Incubator (08/2015 - 12/2015)

Selected Projects

- **Embedded system:** Designed an electronic food dispenser for dogs (software and hardware integration). Worked with technologies such as XBee and Arduino, and developed a web page and an android application to control the system.
- **8-bit Processor:** Designed and described the hardware in VHDL, programmed an assembler in C and implemented it in a Xilinx CPLD.

Accomplishments

- **Distinguished student in Electronics Engineering** (2018)**:** Award that recognizes the best student of the entire degree.
- **Scholarship to study abroad** (2017)**:** Scholarship granted by the university for excellent academic performance.
- **Scholarship for outstanding students** (2015, 2016, 2017)
- **Proyecta 100,000** (2014)**:** Scholarship from Mexican government to study English at a university in the USA.

Coursework

- Programming I and II (C)
- Object Oriented Programming (C++)
- Programming embedded systems (microcontrollers)
- Digital systems II (Computer architectures – VHDL)
- Introduction to data science (Python)
- Complete Python Bootcamp (Udemy)

Skills

- Programming languages: Python (proficient), C (proficient), JavaScript (intermediate), Matlab (intermediate), C++ (prior knowledge)
- Hardware description languages: VHDL and Verilog HDL
- Knowledge of computer architectures and digital systems.
- Data Science (intermediate) and Machine Learning (basic)
- FPGA design and implementation tools: Xilinx and Altera
- Embedded systems and microcontrollers: Arduino, Texas Instruments and Raspberry Pi
- Web development: HTML, CSS and JavaScript (React).
- Lab equipment: Oscilloscope, function generator, multimeter and soldering
- Other: Agile methodology, Git, Linux, Jupyter Notebooks

Languages

- **Spanish** (native)
- **English** (proficient) / TOEFL iBT: 110/120
- **German** (basic)